

Process Synchronization using wait()

```
child is running...
child process terminated.
child completed.
parent process terminated.
aireddy@DESKTOP-HKQRL9G:~/session/session$ nano wait.c
aireddy@DESKTOP-HKQRL9G:~/session/session$ gcc wait.c -o wait
aireddy@DESKTOP-HKQRL9G:~/session/session$ ./wait
Child Process
Child PID = 733
Parent Process
Parent PID = 732
aireddy@DESKTOP-HKQRL9G:~/session/session$ |
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    int pid = fork();
```

```
    if (pid == 0) {
```

```
        printf("Child Process\n");
```

```
        printf("Child PID = %d\n", getpid());
```

```
    }
```

```
    else {
```

```
        wait(NULL);
```

```
        printf("Parent Process\n");
```

```
        printf("Parent PID = %d\n", getpid());
```

```
    }
```

```
    return 0;
```

```
}
```

```
sairedy@DESKTOP-HKQRL9G:~/session/session$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/Program Files/Cloudflare/Cloudflare WARP:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/nodejs:/mnt/c/Users/DELL/AppData/Local/Programs/Python/Python313/Scripts:/mnt/c/Users/DELL/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/DELL/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/Notepad++/IntelliJ IDEA Community Edition/bin:/mnt/c/Users/DELL/AppData/Roaming/npm:/snap/bin
sairedy@DESKTOP-HKQRL9G:~/session/session$ is
fork fork.c process process.c wait wait.c
sairedy@DESKTOP-HKQRL9G:~/session/session$ which gcc
/usr/bin/gcc
sairedy@DESKTOP-HKQRL9G:~/session/session$ name="Ubuntu"
echo $name
Ubuntu
sairedy@DESKTOP-HKQRL9G:~/session/session$ x=10
echo $x
10
sairedy@DESKTOP-HKQRL9G:~/session/session$ echo $abc echo $abc
sairedy@DESKTOP-HKQRL9G:~/session/session$ pwd
/home/sairedy/session/session
sairedy@DESKTOP-HKQRL9G:~/session/session$ cd ..
sairedy@DESKTOP-HKQRL9G:~/session$ echo Hello
Hello
sairedy@DESKTOP-HKQRL9G:~/session$ read name
sairedy@DESKTOP-HKQRL9G:~/session$ export x=10
sairedy@DESKTOP-HKQRL9G:~/session$ unset x
sairedy@DESKTOP-HKQRL9G:~/session$ type echo
echo is a shell builtin
sairedy@DESKTOP-HKQRL9G:~/session$ type echotype echotype echotype echotype ectype echotype echo
-bash: type: echotype: not found
-bash: type: echotype: not found
-bash: type: echotype: not found
-bash: type: echotype: not found
-bash: type: echotype: not found
-bash: type: echotype: not found
echo is a shell builtin
sairedy@DESKTOP-HKQRL9G:~/session$ |
```